

TIME-VARYING TREATMENTS

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Motivation

- Even in the absence of unmeasured confounding factors/model misspecification, standard methods for estimating the causal effect of time-varying treatments on survival are biased when
 - there exists a time-dependent risk factor for survival that also predicts subsequent treatment, and
 - past treatment history predicts subsequent risk factor level

Approach

1. Estimate IP weights

- to create a pseudo-population without confounding by measured variables

2. Conduct a survival analysis in the pseudo-population

- e.g., fit an IP weighted Cox model
 - which estimates the parameters of a marginal structural (Cox) model

1. Estimation of IP weights

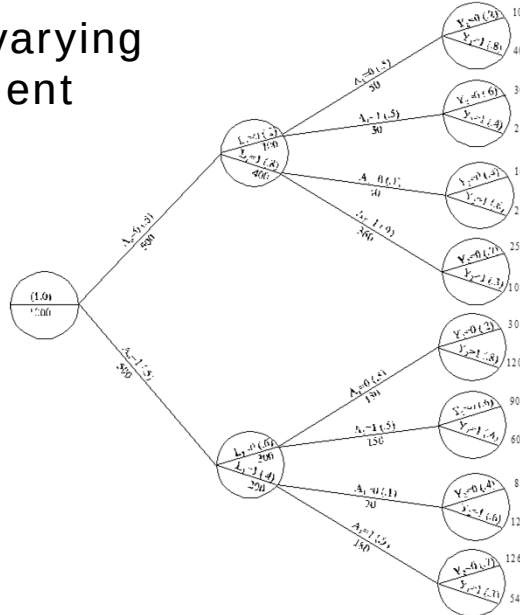
- For time-fixed treatments, the IP weights are time-fixed

- the denominator is, informally, the probability of having one's treatment value given one's confounders values

- For time-varying treatments, the IP weights are time-varying

- the denominator is, informally, the probability of having one's treatment history given one's confounders and treatment history

Time-varying treatment



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Nonstabilized IP weights

□ Time-fixed treatment

$$W^A = \frac{1}{f(A|L)}$$

□ Time-varying treatment

$$W_t^A = \frac{1}{f(A_0|\bar{A}_{-1}, \bar{L}_0)} \times \frac{1}{f(A_1|\bar{A}_0, \bar{L}_1)} \times \dots \times \frac{1}{f(A_t|\bar{A}_{t-1}, \bar{L}_t)} = \prod_{k=0}^t \frac{1}{f(A_k|\bar{A}_{k-1}, \bar{L}_k)}$$

■ Overbar means “history”

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Stabilized IP weights

□ Time-fixed treatment

$$SW^A = \frac{f(A)}{f(A|L)}$$

□ Time-varying treatment

$$SW_t^A = \prod_{k=0}^t \frac{f(A_k | \bar{A}_{k-1})}{f(A_k | \bar{A}_{k-1}, \bar{L}_k)}$$

- Numerator is, informally, the probability of having your own observed treatment history

2. Survival analysis in the pseudo-population

□ For example, fit an IP weighted Cox model with a time-varying treatment but no time-varying covariates

- Each individuals' contribution to each risk set is weighted by SW_t^A
- the parameter estimates of this model are consistent estimates of the parameters of marginal structural Cox model

□ Need to decide which function of the time-varying treatment to include in the model

- Ever use, duration of use, cumulative average dose...

2. Survival analysis in the pseudo-population

- Hazard ratio estimates are sometimes problematic
 - See earlier
- A more informative approach is to estimate the standardized survival curves under hypothetical treatment strategies
 - Always treated, never treated...

An application of a Marginal Structural Cox Model



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Effect of Highly Active Antiretroviral Therapy on Time to Acquired Immunodeficiency Syndrome or Death using Marginal Structural Models

Stephen R. Cole¹, Miguel A. Hernán², James M. Robins^{2,3}, Kathryn Anastos⁴, Joan Chmiel⁵,
Roger Detels⁶, Carolyn Ervin⁷, Joseph Feldman⁸, Ruth Greenblatt⁹, Lawrence Kingsley¹⁰,
Shenghan Lai¹, Mary Young¹¹, Mardge Cohen¹², and Alvaro Muñoz¹

Am J Epidemiol 2003;158:687–694

A real application of MS Cox model

- ☐ MACS
 - Multicenter AIDS Cohort Study
 - 5,622 men (1984-)
- ☐ WIHS
 - Women's Interagency HIV Study
 - 2,628 women (1994-)
- ☐ Semiannual visits
 - questionnaire, blood sample

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MACS+WIHS: Study population

- ☐ Inclusion criteria (1996)
 - HIV-positive
 - AIDS-free
 - Had not started Highly Active Antiretroviral Therapy (HAART)
- ☐ 1,498 participants met inclusion criteria
 - 66% female
 - median age 39 years
 - 37% Caucasians

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MACS+WIHS: Follow-up

☐ Follow-up

- From 1996, or first subsequent eligible visit, to April 2002
- Median follow-up time 5.4 years
- 6,763 person-years of follow-up

☐ Outcome

- Time to AIDS or death
- 329 AIDS + 53 deaths = 382 events
- 259 censored before end of follow-up

Exposure and covariates

☐ Exposure $A(t)$: HAART therapy

- 918 subjects initiated therapy
- Incidence rate: 22/100 person-years

☐ Covariates $L(t)$:

- age, gender, race, prior ART, HIV-1 RNA, CD4, CD8, HIV-related symptoms
- Measured at baseline and every 6 months
- Time-dependent confounders and affected by previous treatment

TABLE 2. Characteristics associated with initiation of highly active antiretroviral therapy for 1,498 human immunodeficiency virus-positive US men and women, 1995–2002

	HR*,†	95% CI*
Male gender	1.11	0.90, 1.35
Age (per 10 years)	0.97	0.88, 1.07
Caucasian race	1.33	1.09, 1.61
CD4 cell count (cells/mm ³)‡		
200–350 vs. >350	1.76	1.47, 2.12
<200 vs. >350	2.26	1.79, 2.85
HIV* RNA level (copies/ml)‡		
401–10,000 vs. <401	2.24	1.81, 2.77
>10,000 vs. <401	2.75	2.22, 3.41
Antiretroviral therapy‡		
Monotherapy vs. none	2.16	1.63, 2.86
Combination therapy vs. none	4.88	4.11, 5.79
PCP* prophylaxis‡	1.14	0.93, 1.39
≥1 HIV-related symptom‡,§	1.04	0.87, 1.24

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TABLE 3. Estimated effect of highly active antiretroviral therapy on time to acquired immunodeficiency syndrome or death for 1,498 human immunodeficiency virus-positive US men and women, 1995–2002

Model and HAART* use	No. of events	Person-years of follow-up	HR*	95% CI*
Unadjusted				
No HAART	238	3,581	1	
HAART	144	3,182	0.98	0.76, 1.26
Adjusted†			0.81	0.61, 1.07
Weighted‡				
No HAART	246	3,586	1	
HAART	125	3,124	0.54	0.38, 0.78‡

* HAART, highly active antiretroviral therapy; HR, hazard ratio; CI, confidence interval.

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Estimation of causal parameter β_1

- ☐ Marginal structural Cox model
 - $\lambda_{Ta}(t) = \lambda_0(t) \exp[\beta_1 a_t]$
- ☐ Fit standard Cox model
 - $\lambda_T[t|\bar{A}_t] = \lambda_{0,T}(t) \exp[\theta_1 A_t]$
- ☐ Reweight subjects in each risk set
 - IP weights have to be estimated (see later)
- ☐ The estimate of θ_1 is an unbiased estimate of the (log) causal hazard ratio β_1

An easy way to handle time-varying IP weights

- ☐ Fit a pooled logistic regression model to approximate Cox regression
$$\text{logit Pr}[D_{t+1}=1|D_t=0, \bar{A}_t] = \theta_{0,t} + \theta_1 A_t$$
 - where $D_t=0$ if subject alive at time t , 1 otherwise
 - θ_1 is a good approximation to β_1 when the probability of $D_t=1$ is small in each t
 - $\theta_{0,t}$ is a time-varying intercept
- ☐ This approximation makes programming simpler

A confidence interval for the causal parameter β_1

- Using weights induces within-subject correlation
 - Naïve variance provided by standard software is incorrect
 - Correct analytical variance not implemented in standard software
- One solution: Use robust variance
 - e.g., sandwich estimator (used in GEE)
 - Robust variance provides conservative confidence intervals for β_1
 - in SAS use Proc Genmod with *repeated* statement rather than Proc Logistic

Estimation of IP weights

$$SW_t^A = \prod_{k=0}^t \frac{f(A_k | \bar{A}_{k-1})}{f(A_k | \bar{A}_{k-1}, \bar{L}_k)}$$

- To estimate factors in denominator, fit a logistic model for
 - $\text{logit Pr}(A_k = 1 | \bar{A}_{k-1}, \bar{L}_k)$
 - Covariates are a function of exposure and covariate history
- Similarly for numerator
 - $\text{logit Pr}(A_k = 1 | \bar{A}_{k-1})$
 - Covariates are a function of exposure history only

Estimation of IP weights

- For a subject in risk set k , multiply the probabilities of having had his/her own treatment history from time 0 to k
 - Use $\Pr[A_k=1]$ for times when $A_k=1$
 - Use $\Pr[A_k=0] = 1-\Pr[A_k=1]$ for times when $A_k=0$
- Similarly for the numerator

What about censoring by loss to follow-up?

- Same strategy: weighting by the inverse of the probability of having one's censoring history
- Compute the stabilized IP weights:

$$SW_t^C = \prod_{k=0}^t \frac{\Pr(C_{k+1} = 0 | \bar{C}_k = \bar{0}, \bar{A}_k)}{\Pr(C_{k+1} = 0 | \bar{C}_k = \bar{0}, \bar{A}_k, \bar{L}_k)}$$

- By fitting two separate logistic models (one for denominator, one for numerator) for censoring
- Final weight at each time t is $SW_t^A \times SW_t^C$

Effect(-measure) modification by baseline covariates

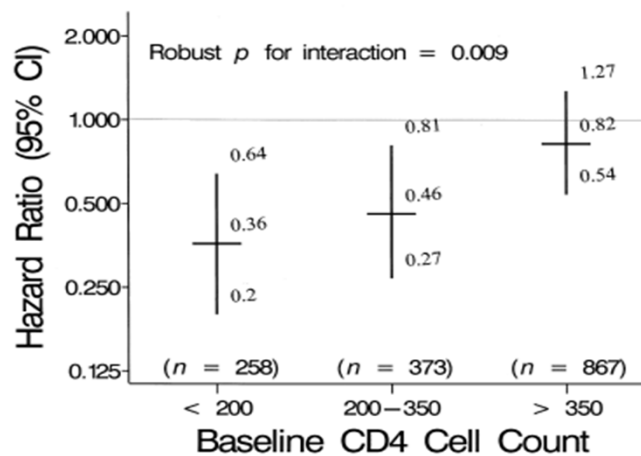
- If interesting from subject-matter standpoint, one can estimate the causal parameter within levels of baseline covariates
 - By fitting the IP weighted model
 - $\text{logit Pr}[D_{t+1}=1|D_t=0, \bar{A}_t] = \theta_{0,t} + \theta_1 A_t + \theta_2' V + \theta_3' V \times A_t$
 - V is a subset of L_0
- Not to adjust for confounding by V
- IP weights are redefined as

$$SW_t^A(V) = \prod_{k=0}^t \frac{f(A_k | \bar{A}_{k-1}, V)}{f(A_k | \bar{A}_{k-1}, \bar{L}_k)}$$

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In our study



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MSMs

☐ Advantages

- Resemble standard models
- Any type of outcome variable

☐ Disadvantages

- Less useful to estimate treatment interaction with time-varying covariates
- Inference sensitive to extreme weights

Readings

- ☐ Robins JM, Hernán MA. Estimation of the causal effects of time-varying exposures.
 - In: Longitudinal Data Analysis. Fitzmaurice G, Davidian M, Verbeke G, Molenberghs G, eds. New York: Chapman and Hall/CRC Press, 2008, pp. 553-599.
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- ☐ Hernán MA, Brumback B, Robins JM. Marginal structural models to estimate the causal effect of zidovudine on the survival of HIV-positive men. *Epidemiology* 2000; 11(5):561–570
- ☐ HIV-CAUSAL Collaboration. The effect of combined antiretroviral therapy on the overall mortality of HIV-infected individuals. *AIDS* 2010; 24(1):123-137.